

NITHYASRI SRIKUMAR

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science
Master of Science in Engineering in Computer and Information Science

Philadelphia, PA
Expected May 2027

SRM Institute of Science and Technology, College of Engineering
Bachelor of Technology in Computer Science and Engineering
Cumulative GPA: 9.73/10.00 (scale: 0.00-10.00)

Chennai, India
May 2023

SKILLS

Languages: Python, C, C++, Java, JavaScript, R, SQL

Frameworks and databases: React, Spring Boot, Express, Flask, PostgreSQL, MongoDB, Firebase, DB2

Libraries: Pandas, NumPy, Matplotlib, OpenCV, Scikit-Learn, Librosa, TensorFlow, PyTorch, Hugging Face Transformers, Tokenizers, NLTK

Technical Domains: Machine Learning, Reinforcement Learning, Deep Learning, NLP, Computer Vision, Explainable AI, Data Structures & Algorithms

RESEARCH EXPERIENCE

Research Assistant, xLab, University of Pennsylvania, Sep 2025 – Present

Philadelphia, PA

Supervisor: Prof. Rahul Mangharam

- Implemented safe reinforcement learning combining Hamilton-Jacobi reachability analysis with PPO, achieving 40% reduction in safety violations in autonomous navigation tasks while maintaining task performance.
- Developed safety Q-network using PyTorch that predicts constraint violation costs and applied safety shielding by filtering unsafe actions before execution.
- Tested and validated algorithms in multi-agent autonomous vehicle simulation environments, demonstrating practical application to real-world safety-critical systems,

Multimodal Emotion Recognition from Audio and Video using CNN, Apr 2023

Chennai, India

- Achieved multimodal emotion recognition from audio and video, improving accuracy by over 60% compared to unimodal models, by combining audio features (STFT, MFCC via Librosa) and video features (facial landmarks with OpenCV) and applying CNNs with decision-level fusion.
- Demonstrated robust emotion classification, through a live application with real-world use in conversational agents, recommendation systems, and smart environments, validated through precision, recall, and F1-score, by integrating CNN-based facial expression models and systematically evaluating multimodal predictions.

PROFESSIONAL EXPERIENCE

Wells Fargo, Software Engineer, Aug 2023 – Jul 2025

Bengaluru, India

- Modernized legacy mainframe systems into cloud-ready web apps, decommissioning three monolithic services and reducing infrastructure costs by 20%, through React.js (Redux, hooks) front end and Spring Boot microservices.
- Improved client-side performance by 30%, validated via Lighthouse and Core Web Vitals, by reducing React bundle size 40%, implementing code-splitting, and optimizing state management.
- Enhanced backend stability and maintainability, lowering production incidents by 25% over a year, by re-engineering business logic into modular Java services, upgrading React (16→18) and Java (11→17), and improving code quality via SonarQube.
- Implemented secure enterprise authentication with SSO, reducing auth-related defects by 40%, by integrating Ping Identity, OAuth2, JWT, and role-based access control (RBAC).

Wells Fargo, Summer Intern, Jun 2022 – Aug 2022

Virtual

- Built and deployed a stock market charting web app with daily active users, using React.js, Node.js, Firebase, and custom APIs as a key differentiator.
- Reduced data latency and optimized server load by 20%, validated via real-time benchmarks, by implementing dynamic polling intervals and real-time data fetching with Node.js and Firebase.

- Enhanced user engagement and insights, measured by increased watchlist interactions and chart views, by developing personalized watchlists and interactive market trend visualizations using Chart.js.
- Delivered predictive investment insights, demonstrated through successful stock price forecasts in production, by integrating LSTM models using Python (Keras, TensorFlow) and visualizing predictions on the React front end.

PROJECTS

Retrieval-Augmented QA System with Transformer Models, *Independent Project*, Jun 2024 – Aug 2024

- Built production-ready context-aware QA system, demonstrated through accurate answer extraction from retrieved passages, by combining dense retrieval (Sentence Transformers + FAISS) with DeBERTa extractive QA, orchestrated via LangChain.
- Overcame transformer context limitations, validated through successful retrieval from large corpora, by implementing RAG with scalable FAISS nearest-neighbor search.

Style Transfer using CycleGANs, *Team of 2*, Sep 2022 – Dec 2022

- Achieved high-quality artistic style transfer on images, measured by visually enhanced outputs, by developing and optimizing CycleGAN generator/discriminator networks with perceptual loss and instance normalization.

Collusion Detection in Auctions Using Clustering & Statistical Analysis, *Academic Project*, Jan 2022 – May 2022

- Detected dynamic collusion and enhanced auction fairness by identifying anomalous bidder groups and bid rotations using DBSCAN clustering, Markov Chains for sequence modeling, and Z-Score statistical analysis.

ACHIEVEMENTS

Performance Based Scholarship, May 2020

Chennai, India

- Performance-based scholarship of Rs. 50,000 for securing the first rank (10 GPA) in the university examinations conducted in Year I.

CERTIFICATIONS AND TRAINING

- Machine Learning — Stanford Online (Coursera)
- Fundamentals of Deep Learning — NVIDIA
- Natural Language Processing with Probabilistic Models — DeepLearning.AI
- Build Generative Adversarial Networks (GANs) — DeepLearning.AI